

Project Design Phase
Proposed Solution Template

Date	15 February 2025
Team ID	LTVIP2025TMID36086
Project Name	Enchanted Wings: Marvels of Butterfly Species
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Identifying butterfly species in the field is time-consuming, requires expert knowledge, and lacks accurate, real-time classification tools accessible to all.
2.	Idea / Solution description	Develop a butterfly image classification system using transfer learning (e.g., VGG16). Users can upload an image and instantly receive species identification.
3.	Novelty / Uniqueness	Trained on a custom dataset of 75 butterfly species (6499 images), this solution combines deep learning, real-time prediction, and educational features in one.
4.	Social Impact / Customer Satisfaction	Enables field researchers, educators, and citizen scientists to contribute to biodiversity monitoring. Encourages youth in environmental learning and research.
5.	Business Model (Revenue Model)	Freemium model: basic identification tool is free. Premium version includes offline access, species insights, contributions tracking, and institutional licensing.
6.	Scalability of the Solution	Can be extended to other insect or animal species (e.g., birds, moths, beetles). Supports multilingual UI and mobile integration for wide adoption.

1. Problem Statement (Problem to be solved)

Identifying butterfly species in the field is often time-consuming, error-prone, and requires expert-level knowledge. Existing solutions are either inaccurate, limited in scope, or not accessible for use in diverse environments by students, researchers, or citizen scientists.

2. Idea / Solution Description

We propose an AI-powered butterfly image classification system that uses transfer learning techniques (VGG16) to accurately identify butterfly species from uploaded images. The system is trained on 6499 images across 75 classes and deployed using a user-friendly interface (Streamlit) for real-time predictions and educational insights.

3. Novelty / Uniqueness

- Combines computer vision with biodiversity conservation.
 - Trained specifically on 75 butterfly species—tailored to real-world field requirements.
 - Integrates prediction with learning, giving not just the species name but also related biological context.
 - Designed to be used by both professionals and the general public with no technical background.
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4. Social Impact / Customer Satisfaction

- Supports biodiversity conservation by aiding in species monitoring and habitat analysis.
 - Empowers students and citizen scientists to engage in research and environmental education.
 - Saves time for researchers and provides immediate, reliable identification assistance.
 - Builds environmental awareness and fosters interest in ecology and machine learning.
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5. Business Model (Revenue Model)

- **Freemium Model:**
Basic species identification is offered free to the public.
A premium version includes features such as:
 - Offline access
 - High-resolution recognition
 - Expert species data

- Institutional use licensing for research bodies and universities

6. Scalability of the Solution

- Can be extended to classify other insects (e.g., moths, beetles) or even birds and plants.
- Expandable dataset and model retraining pipeline enable continuous improvement.
- The interface can be localized into multiple languages for wider global use.
- Suitable for deployment as a mobile app or integration with camera drones in field research.